

LINEAR PROGRAMMING

FOOD INDUSTRY PRODUCTION OPTIMIZATION

**Maximizing Profit using Graphical Method in Linear
Programming
Food Production Optimization**



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INTRODUCTION

In the food industry, companies always aim to maximize their profit while using limited resources efficiently. These resources can include cooking time, labor, and equipment capacity.

Linear Programming is a mathematical method that helps decision-makers determine the best way to allocate these limited resources. It allows companies to decide how much of each product to produce in order to achieve the highest possible profit.

In this presentation, we will apply Linear Programming to a food production problem involving two products. We will formulate the mathematical model, define the objective function and constraints, and solve the problem using the graphical method.

THE REAL WORLD PROBLEM

A food company produces two types of meals:

Chicken Meals & Beef Meals

The objective is to maximize the total profit while adhering to available resource constraints such as cooking time and station capacity, along with non-negativity constraints.

DECISION VARIABLES AND OBJECTIVE FUNCTION

Decision Variables:

X1 = Number of Chicken Meals produced (units)

X2 = Number of Beef Meals produced (units)

Objective Function:

$$\text{MAX } Z = 60X_1 + 90X_2$$

Max	▼	Z =	60	X ₁ +	90	X ₂
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Subject to constraints

2	X ₁ +	6	X ₂	≤	▼	30
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5	X ₁ +	3	X ₂	≤	▼	80
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and x₁, x₂ ≥ 0

RESOURCE CONSTRAINTS AND REQUIREMENTS

Production Requirements:

Chicken Meals (X1): require 2 hours on Cooking Station 1 and 5 hours on Cooking Station 2

Beef Meals (X2): require 6 hours on Cooking Station 1 and 3 hours on Cooking Station 2

Available Resources Daily:

Cooking Station 1: 30 hours maximum

Cooking Station 2: 80 hours maximum

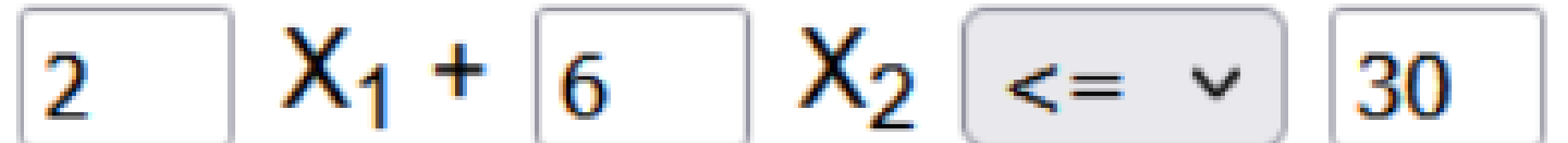


MATHEMATICAL CONSTRAINTS

Cooking Station 1 Constraint:

$$2X_1 + 6X_2 \leq 30$$

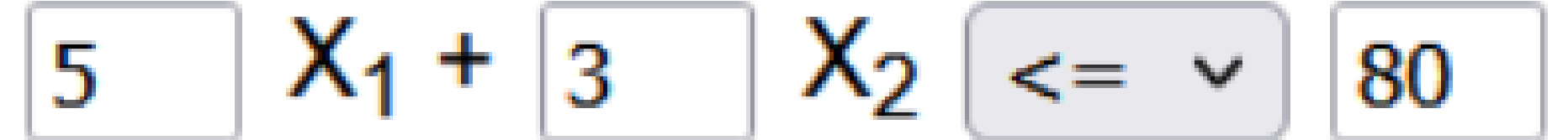


A graphical representation of the first constraint. It consists of a sequence of elements: a text box containing the number '2', the variable 'X₁', a '+' sign, a text box containing the number '6', the variable 'X₂', a grey button containing '<= v', and a text box containing the number '30'. Each text box and the button have a thin grey border.

Cooking Station 2 Constraint:

$$5X_1 + 3X_2 \leq 80$$



A graphical representation of the second constraint. It consists of a sequence of elements: a text box containing the number '5', the variable 'X₁', a '+' sign, a text box containing the number '3', the variable 'X₂', a grey button containing '<= v', and a text box containing the number '80'. Each text box and the button have a thin grey border.

Non-negativity Constraint:

$$X_1, X_2 \geq 0$$



A graphical representation of the non-negativity constraint. It is a single text box with a thin grey border containing the text 'and x1,x2 >= 0'.

COMPLETE LINEAR PROGRAMMING MODEL

$$\text{MAX } Z = 60X_1 + 90X_2$$

Subject to:

$$2X_1 + 6X_2 \leq 30 \text{ (Cooking Station 1)}$$

$$5X_1 + 3X_2 \leq 80 \text{ (Cooking Station 2)}$$

$$\text{and } X_1, X_2 \geq 0$$



SOLUTION METHOD - 1

Graphical Method

Steps for Solving:

To graph the constraint, we first convert the inequality into an equation. Then, we calculate the intercept points by setting one variable equal to zero. These points are used to draw the constraint line on the graph.

Treat it as $2x_1 + 6x_2 = 30$

When $x_1 = 0$ then $x_2 = ?$

$$\Rightarrow 2(0) + 6x_2 = 30$$

$$\Rightarrow 6x_2 = 30$$

$$\Rightarrow x_2 = \frac{30}{6} = 5$$

When $x_2 = 0$ then $x_1 = ?$

$$\Rightarrow 2x_1 + 6(0) = 30$$

$$\Rightarrow 2x_1 = 30$$

$$\Rightarrow x_1 = \frac{30}{2} = 15$$

x_1	0	15
x_2	5	0

SOLUTION METHOD - 2

Graphical Method

Steps for Solving:

For the second constraint, we also convert the inequality into an equation. Then, we calculate the intercept points by setting one variable equal to zero each time. The resulting points help us draw the second constraint line accurately on the graph.

Treat it as $5x_1 + 3x_2 = 80$

When $x_1 = 0$ then $x_2 = ?$

$$\Rightarrow 5(0) + 3x_2 = 80$$

$$\Rightarrow 3x_2 = 80$$

$$\Rightarrow x_2 = \frac{80}{3} = 26.67$$

When $x_2 = 0$ then $x_1 = ?$

$$\Rightarrow 5x_1 + 3(0) = 80$$

$$\Rightarrow 5x_1 = 80$$

$$\Rightarrow x_1 = \frac{80}{5} = 16$$

x_1	0	16
x_2	26.67	0

Graphical Method



Find optimal solution

The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 60x_1 + 90x_2$
$O(0, 0)$	$3 \rightarrow x_1 \geq 0$ $4 \rightarrow x_2 \geq 0$	$60(0) + 90(0) = 0$
$A(15, 0)$	$1 \rightarrow 2x_1 + 6x_2 \leq 30$ $4 \rightarrow x_2 \geq 0$	$60(15) + 90(0) = 900$
$B(0, 5)$	$1 \rightarrow 2x_1 + 6x_2 \leq 30$ $3 \rightarrow x_1 \geq 0$	$60(0) + 90(5) = 450$

Final Result

Optimal Solution

Decision Variable Values:

$X_1 = 15$ units (Chicken Meals)

$X_2 = 0$ units (Beef Meals)

Maximum Total Profit:

Max $Z = \$900$

The maximum value of the objective function $Z = 900$ occurs at the extreme point $(15, 0)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 15, x_2 = 0$ and $\max Z = 900$.

INTERPRETATION OF RESULTS

The solution indicates that the maximum profit is achieved when producing 15 units of Chicken Meals and 0 units of Beef Meals.

This means:

Cooking Station 1 resources are fully utilized

The company achieves the highest possible profit

The maximum total profit is \$900



CONCLUSION

The application of Linear Programming using the Graphical Method provides an effective tool for improving decision-making and maximizing profit in the food industry.

This optimization approach helps managers determine the best production plan, use cooking stations efficiently, and make better use of available resources.

The same mathematical model can also be applied to many real-world problems such as production planning, scheduling, inventory management, and resource allocation in different industries.

THANK YOU.